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ASSESSING THE BENEFITS OF OPTIMIZED AGENTIC AI SYSTEMS FOR ASSET PRICING

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ABSTRACT

Evaluating optimized AI systems for asset pricing is fundamentally difficult for two reasons. First, models are trained on all data, implying that any backtest or analysis using historical data suffers from look-ahead bias. In addition, markets are reflexive — as investors adopt AI, prices adjust — which may erode the very patterns the AI system was trained to exploit. We introduce a real-time, out-of-sample benchmark designed to sidestep both problems. The benchmark measures how well AI systems can explain contemporaneous stock returns around earnings announcements using only information available at announcement time, including the text of the announcement itself. Applying this benchmark to a range of agentic AI systems — which extract structured signals from earnings call transcripts and optimize over those signals — we find that the best-optimized systems more than double the explained variation in returns relative to standard benchmarks (R2 increasing from 8% to close to 20%). We show that AI-based optimization can deliver efficiency gains relative to traditional machine learning methods while also improving interpretability as our approach produces human-readable economic mechanisms that explain price movements. These learned rules can be compared to the drivers of realized returns in existing asset pricing models to identify missing sources of variation in a data-driven, self-evolving way that integrates empirical learning with economic structure. We release an SDK for researchers to improve on our results. Saturating this benchmark would represent fundamental progress in understanding how capital markets process firm-level information.

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1 Introduction

Modern AI systems have quickly become more capable across a wide range of domains. This naturally raises the question whether these models can also be used to analyze financial markets and explain asset prices. The apparent random movements in asset prices has long puzzled financial economists. The challenge is perhaps most salient around earnings releases. During a short window, stock prices display large moves which have proved difficult to explain, e.g., earnings surprises (measured in various ways) explain up to about 5% of the variation in returns (see, e.g., Chiang et al., 2019; Hand et al., 2022). It is therefore natural to ask how well modern AI models can explain this variation and reduce the "dark matter" of stock price fluctuations.

Using AI systems to explain stock price movements faces at least two challenges. First, it is well known that large language models (LLMs) suffer from look-ahead bias as they have been trained on vast amounts of data, including data on returns. While attempts have been made to mitigate the impact of look-ahead bias (e.g., Glasserman and Lin, 2023; Engelberg et al., 2025; Merchant and Levy, 2025), look-ahead bias is unlikely to be fully addressed unless the models are trained with hard knowledge cutoffs. A possible solution is to train generative models with knowledge cutoffs (Drinkall et al., 2024; Levy, 2024; He et al., 2025), but those models significantly underperform frontier AI models and can at best provide a lower bound on the ability of optimized AI systems to explain asset prices. Moreover, a growing literature highlights the benefits of *optimizing* AI systems, i.e., by optimizing prompts, context, or updating the model's parameters via supervised finetuning (SFT) or reinforcement learning (RL). If the AI model has knowledge of the announcement return, optimization is more likely to recover the information from the model and increase look-ahead bias (Hu et al., 2024).

Second, while natural language processing has long been used by investors, modern AI models have significantly increased the set of tools available to market participants. As a result, in equilibrium, it is plausible that the relationship between text and returns evolves as the model capabilities expand and investors find new ways to deploy AI. As we are initially primarily interested in how well optimized AI systems explain current prices, conclusions based on historical data may not carry over to the current market environment. Understanding the evolution and adoption of AI systems is equally interesting, but this requires real-time performance measurement as a key input for the reasons mentioned before.

To address these challenges, we make three contributions. First, we introduce a new benchmark that evaluates optimized AI systems out-of-sample (OOS) in real time to explain stock price movements. The inputs to the model, which include the context and prompts that

the AI model uses to make its forecasts (as well as the model parameters), are determined prior to the earnings announcement. We then feed the model the earnings call information (i.e., the earnings numbers as well as the earnings call transcript) and let it predict the announcement return.

Our second contribution is to consider a broad range of methods and models to assess their performance during the last quarter of 2025 on our new benchmark. Future versions of this paper will extend these results to subsequent earnings quarters. We provide three main empirical findings. First, frontier large language models (LLMs) can explain up to 15% of the variation in returns based on the earnings call transcript and earnings surprise information alone, which is significantly higher than the prior literature that explains up to 8% of the return variation (Hand et al., 2022). Second, this average number masks a significant amount of inter-industry variation. While there is little heterogeneity in the ability of earnings surprises to explain returns across industries, optimized AI systems perform much better in some industries than in others. We also find that no one model dominates; one model may outperform for financial firms, while another model outperforms in utilities. Third, there are non-trivial benefits from optimizing model prompts and context. We focus our attention on exploring two state-of-the-art, gradient-free methods of prompt optimization: Genetic-Pareto (GEPA) search (Agrawal et al., 2026) and Agentic Context Engineering (ACE) (Zhang et al., 2026). Both methods follow an iterative process on a training sample to develop a prompt or additional context that guides the model's reasoning to predict the contemporaneous return. We then evaluate the optimized prompts and context out-of-sample and find that it raises the fraction of explained variation to close to 20%.

Our third contribution is to show how these optimization methods provide useful, interpretable economic insights. For instance, the optimization results reveal the economic mistakes made by unoptimized AI systems. GEPA teaches the model that "good" is not the same as "news." In the default prompt, the model tends to predict positive announcement returns for firms with strong operating performance, record profits, revenue growth, or buybacks. The optimized prompt instead learns that public firms are priced relative to expectations: a company can report strong results and still have a neutral or negative announcement return if those results were already embedded in the stock price. The sector-specific rules sharpen this point. For utilities and telecommunications firms, the market places more weight on churn and free cash flow than subscriber growth; for high-growth technology firms, distant product roadmaps are discounted relative to immediate financial delivery; and for financial firms, buybacks are often treated as routine capital maintenance rather than a new positive signal. In this sense, GEPA recovers a simple but central asset-pricing principle: announcement returns are driven by belief revisions, not by unconditional

performance.

ACE produces richer economic mechanisms because the learned rulebook captures interactions among fundamentals, guidance, and the market's prior beliefs. The rules indicate that markets penalize cautionary forward-looking language that lowers the path of expected cash flows and headline earnings or margins inflated by transitory items. They also show that negative operational information can dominate otherwise positive narratives: record margins receive little credit when they come with one-time benefits, execution failures, or weakness in an important segment. These patterns map naturally into familiar economic concepts, including the distinction between permanent and transitory earnings, the role of management guidance in belief formation, and industry-specific thresholds for what counts as value-relevant news. An implication is that optimized agentic systems generate candidate determinants of realized returns that can be compared to the drivers of realized returns in asset pricing models.

More broadly, our results illustrate that optimized AI systems have an important role to play in guiding future empirical and theoretical research in asset pricing by generating interpretable explanations of realized asset price movements—notably these explanations hold out-of-sample. We believe that this is a key benefit of AI compared to traditional machine learning (ML) methods. For example, traditional statistical methods are agnostic to the labeling of the data, i.e., predictions would be unaffected by randomizing the column labels in the data. In contrast, AI systems benefit from pretraining on vast amounts of data which can then be used to understand the nature of variables and how they are connected. This pretraining serves as an economic prior and enables more rapid learning through writing rulebooks that generalize across calls. This logic extends to many other questions in finance and economics more broadly. To the best of our knowledge, our paper is the first to show that this hypothesis holds out-of-sample in financial markets.

We do emphasize that, although we apply recently-developed methods to frontier LLMs and find significant gains already, we think that we have only scratched the surface and that further improvements are within reach. For instance, we have not used information (i) from earlier earnings calls (e.g., some CEOs may generally be more optimistic than others), (ii) from peers' earnings calls or (iii) additional information that is typically available right before a call to pinpoint what specific items analysts are looking for in the upcoming call. Moreover, model capabilities are expanding rapidly as are the prompt and context optimization methods. This makes for a clear and exciting journey to discovering what moves asset prices. Also, the benchmark naturally refreshes itself as the topics and themes that drive markets (e.g., from the great financial crisis to the European sovereign debt crisis or the COVID-19 pandemic) change and this may modify the connection between text and

returns. In the ultimate case where the benchmark gets saturated, as has been the case with many other AI benchmarks, we have made great strides in understanding how investors price key information events.

To simplify and coordinate the effort to discover how optimized AI system can be used to explain asset prices, we will soon release an SDK and launch an open competition inviting researchers to improve on our results on explainingmarkets.ai.

Related literature Our paper builds on a vast literature showing that earnings announcements convey value-relevant information. The empirical foundation begins with Beaver (1968), who shows that accounting earnings announcements are associated with movements in stock prices, trading volume, and investor belief revision, thereby making earnings releases a natural setting for studying how markets process firm-level information. Related, Ball and Brown (1968) show that while the market prices a large portion of the information content in an earning announcement ex ante, the market tends to under-react on average, i.e., the prices exhibit "post-earnings announcement drift." Bernard and Thomas (1989) and Bernard and Thomas (1990) extend this literature by showing that delayed price responses are linked to the implications of current earnings for future earnings, suggesting limits to immediate market processing even for public, salient disclosures. Kothari (2001) synthesizes this broader capital-markets research agenda around value relevance, market efficiency, fundamental analysis, and the interpretation of accounting information in asset prices. Our paper also relates to a long literature in asset pricing, building on the seminal work of Fama et al. (1969), to explain realized movements in stock returns. Relative to this literature, our setting keeps the earnings-announcement event at the center but asks whether modern AI systems can explain contemporaneous announcement returns using only information available at announcement time, rather than focusing primarily on delayed drift or on a fixed accounting signal.

The paper also relates to research showing that qualitative and nonnumerical disclosure contains information beyond headline earnings numbers. Tetlock (2007) demonstrates that media pessimism extracted from financial news predicts market returns and trading volume, linking textual tone to investor sentiment and price pressure. Loughran and McDonald (2011) show that dictionary-based sentiment measures must be adapted to financial language, because generic negative-word lists misclassify common business terms and can distort inference in financial applications. Mayew and Venkatachalam (2012) extend the disclosure literature beyond written text by showing that managers' vocal affect during earnings conference calls is informative about future firm performance and is incremental to both quantitative earnings information and linguistic content. These papers motivate the

view that market participants respond not only to numerical earnings surprises but also to softer information embedded in narratives, tone, language, and delivery.

Our contribution is to move from handcrafted disclosure measures to optimized agentic AI systems that extract and combine signals from earnings-announcement information in real time. Because frontier models may have been trained on historical outcomes, historical backtests of AI performance in asset pricing can suffer from look-ahead bias; and because markets are reflexive, relationships between disclosures and returns may evolve as investors adopt new AI tools. We therefore introduce a real-time, out-of-sample benchmark for evaluating how well optimized AI systems explain announcement-window returns using only contemporaneously available information, including earnings-call transcripts and announcement data. Moreover, the AI systems generate interpretable, human-readable mechanisms that can be evaluated out of sample and compared with the drivers of realized returns in asset pricing models, thereby linking the earnings-announcement and soft-disclosure literatures to a broader question of how modern AI can help empirical measurement of information processing in financial markets.

2 Benchmarking Agentic AI Systems

Evaluating the extent to which AI systems can explain asset prices requires a benchmark that is both methodologically rigorous and economically meaningful, e.g., free of look-ahead bias and focused on a task that is relevant. In this section, we outline the desirable features of such a benchmark, formally define its structure, and explain our approach to out-of-sample evaluation.

2.1 Desirable Features of a Benchmark

The application of LLMs, agentic tooling, and AI models in a more general sense to a variety of domains has shown rapid progress (see, e.g., Jumper et al., 2021). Moreover, a growing literature highlights the benefits of optimizing AI systems by modifying prompts, context, or model parameters (Ouyang et al., 2022; Khattab et al., 2024; Agrawal et al., 2026). Although a variety of domains have seen rapid progress, it is unclear whether we should expect similar benefits when applying these methods to tasks involving financial markets.

For example, the fact that long time-series of content are used to train these models implies that the models will exhibit look-ahead bias for many in-sample tasks. Layering optimization on top of such models and systems is likely to exacerbate the bias. Even if one ignores look-ahead bias, the reflexivity of markets implies that the benefits of the technology

may not hold out of sample, i.e., the value of information or technology today differs from its value tomorrow if investors adjust their models and strategies in response to profitable trading opportunities. Thus, empirical research involving AI systems needs to adapt, and studying the adaptation of markets to new AI methods requires measurement on a real-time out-of-sample benchmark.

Acknowledging the differences and challenges associated with empirical research involving AI, we first outline the desirable features of *any* benchmark:

1. Construct Validity - the benchmark measures what it claims to measure.
2. Contamination Resistance - "answers" should not be readily available due to leakage at training or inference time.
3. Ecological Validity - alignment with real-world task performance.
4. Discriminative Power - systems with different capabilities achieve meaningfully different scores.
5. Reproducibility & Transparency - benchmark is fully and openly specified.

Guided by these features, we design a benchmark that examines the extent to which systems can explain the consequences of information releases on asset prices. As we will show in the next section, when such a benchmark is evaluated in real-time, it naturally satisfies these features.

2.2 Benchmark Definition

We formalize the benchmark within the context of a standard event study framework. Consider a stylized factor model where the returns of asset i around time t over a horizon h are given by:

$$r_{i,[t,t+h]} = \boldsymbol{\beta}_{i,t}^{\top} \mathbf{F}_{[t,t+h]} + \epsilon_{i,[t,t+h]}. \tag{1}$$

The benchmark isolates cases where content $\omega_{i,t}$ relevant to asset i is released at time t. Using an earnings announcement as an example, asset i represents a firm's equity, and $\omega_{i,t}$ represents the corpus of text and audio released during the event. The idiosyncratic return component can thus be decomposed as:

$$\epsilon_{i,[t,t+h]} = g(\omega_{i,t}, \Omega_{t-}) + \nu_{i,[t,t+h]} \tag{2}$$

where Ω_{t-} is the investors' information set strictly prior to the release, $g(\cdot)$ is the unobservable market-wide information pricing function, and $\nu_{i,[t,t+h]}$ is orthogonal noise. During key event windows, such as earnings releases, we assume that the variance of $\nu_{i,[t,t+h]}$ is typically small relative to the variance of $g(\cdot)$.[1]

Given this setup, the objective is to measure how well an AI system can approximate the true pricing function $g(\cdot)$. Formally, let $\mathcal{E} = \{(i_k, t_k)\}_{k=1}^{N}$ denote the benchmark evaluation sample of discrete information events N. A valid submission to the benchmark consists of a set of scalar predictions, $\{\hat{g}(\omega_{i,t}, \Omega_{t-})\}_{(i,t)\in\mathcal{E}}$, generated by the AI system. In practice, the AI system is unlikely to have all the information that is relevant to investors and will therefore be conditional on $\omega_{i,t}$ and only a subset of Ω_{t-}. As AI systems can expand the information set by incorporating information in earlier calls of the same company, information in earnings calls of related firms, and news announcements that occur between earnings calls, better predictions can be due to improved understanding of $\omega_{i,t}$ or a more accurate approximation of Ω_{t-}.

Because both $g(\cdot)$ and ν are unobservable, we evaluate the performance of a submission based on its explanatory power in a cross-sectional regression:

$$\epsilon_{i,[t,t+h]} = \alpha + \gamma\hat{g}(\omega_{i,t}, \Omega_{t-}) + \boldsymbol{\theta}^{\top}\mathbf{X}_{i,t} + u_{i,[t,t+h]}, \tag{3}$$

where $\epsilon_{i,[t,t+h]}$ is the observed residual from the factor model, $\mathbf{X}_{i,t}$ is a vector of known, quantitative determinants of the idiosyncratic market reaction (such as the standardized unexpected earnings, or SUE), and $u_{i,[t,t+h]}$ is the regression residual.

The primary performance statistic is the incremental explanatory power, or ΔR^2, generated by the AI system's predictions over a baseline model, e.g., one that includes only the quantitative controls $\mathbf{X}_{i,t}$ or the current "best" system. This relates the AI's edge directly to the inherent noise in the system. Assuming g and ν are orthogonal, the theoretical maximum R^2 a perfectly omniscient system could achieve is bounded by:

$$\overline{R}^2 = \frac{Var(g)}{Var(g) + Var(\nu)}. \tag{4}$$

This benchmark design is general enough to be of broad interest to both academics and investment professionals. Because it closely mirrors research designs used to study macroeconomic announcements and corporate events, it can be extended easily to those settings, which we are exploring in ongoing work.

[1] In principle, $\omega_{i,t}$ can also contain information about factor loadings or realizations, (e.g., for bellwether firms moving the market or certain factors). By definition, we measure the net effect of the factors in this case.

2.3 A First Implementation: Earnings Announcements

Our implementation of the benchmark focuses on corporate earnings announcements (EAs) for three primary reasons. First, a rich literature spanning more than five decades has sought to explain the returns surrounding earnings announcements. Progress from the earliest iterations to today has been an increase in explanatory power from roughly 2% to 8%, with much of this coming from more robust measurement (Chiang et al., 2019) or from incorporating numerous surprises along dimensions other than just earnings (Hand et al., 2022). This baseline suggests that the benchmark will either demonstrate significant discriminative power across AI models, or reveal that asset prices around these events are overwhelmingly driven by unobservable noise that cannot be interpreted using current AI models.

Second, EAs are information-rich events involving content across a variety of modalities: text, audio, and sometimes visual presentations. As such, the market reaction is likely a function of both quantitative and qualitative information. This multi-modal synthesis is precisely where LLMs and AI agents are expected to add significant value beyond what historical, lexicon-based methods have achieved.

Third, there are more than 1,000 EAs every quarter. Even assuming moderate effect sizes, this yields a sufficiently large sample to provide statistically powerful, real-time tests of performance differences across systems, while being highly confident that the benchmark is robust against data contamination.

2.3.1 Event Date and Return Measurement

The core of our benchmark relies on isolating the market reaction to the specific content released during the EA. We measure the abnormal return in a narrow one-day window (i.e., we set $h = 1$) strictly aligned with when the information becomes actionable for market participants. For an earnings event that occurs at time t_c on the calendar date d, the effective trading event date t is defined as:

$$t = \begin{cases} d & \text{if } t_c \leq 16:00 \text{ ET and } d \text{ is a trading day} \\ \tau(d) & \text{if } t_c > 16:00 \text{ ET or } d \text{ is a non-trading day} \end{cases} \tag{5}$$

where $\tau(d)$ denotes the next trading day available following the calendar date d.

Abnormal returns are measured using a standard market-adjusted return. For stock i on trading day t, we define the abnormal return as:

$$r_{i,t}^{abn} = r_{i,t} - r_{m,t}, \tag{6}$$

where $r_{i,t}$ is the total stock return (including dividends) and $r_{m,t}$ is the return on the broader market. This market-adjusted return serves as our empirical proxy for the idiosyncratic residual $\epsilon_{i,[t,t+h]}$ defined in Equation (1). To enable real-time out-of-sample measurement, we proxy the market return using the Vanguard Total Stock Market ETF (VTI). Both returns are calculated using close-to-close prices from $t-1$ to t.

2.3.2 Earnings Surprise Baseline

A critical question is how well existing approaches explain announcement returns. The standard measure in the literature is the earnings surprise: the difference between reported earnings and analyst consensus expectations formed strictly prior to the announcement. Recalling our general control vector $\mathbf{X}_{i,t}$, we define its primary component, the standardized earnings surprise $s_{i,t}$, for an announcement at time t as:

$$s_{i,t} = \frac{e_{i,t} - \mathbb{E}[e_{i,t}|\Omega^a_{t-}]}{p_{i,t-}}, \tag{7}$$

where $e_{i,t}$ is the reported earnings per share, Ω^a_{t-} is the analysts' information set, $\mathbb{E}[e_{i,t}|\Omega^a_{t-}]$ is the consensus (mean) analyst forecast formed prior to the release, and $p_{i,t-}$ is the pre-announcement stock price.

2.3.3 Cross-Sectional Percentile Rank

To ensure comparability across different time periods and varying market regimes, we convert the abnormal return and the earnings surprise to cross-sectional percentile ranks. Within each distinct evaluation period q (e.g., a calendar quarter), we compute the percentile rank as:

$$PRank(x_{i,t}) = \frac{\text{rank}(x_{i,t})}{N_q}, \tag{8}$$

where $\text{rank}(x_{i,t})$ assigns a value of 1 to the most negative value of x and N_q to the most positive, and N_q is the total number of valid observations in evaluation period q. Henceforth, when we say abnormal return or earnings surprise, we are referring to the percentile ranked version of the variable, i.e., $PRank(r^{abn}_{i,t})$ and $PRank(s_{i,t})$, respectively, rather than the raw value.

We prefer to use percentile ranks as opposed to the actual return or earnings surprise to curb the impact of outliers.

2.3.4 Real-Time Out-of-Sample Evaluation

We follow a strict temporal protocol when evaluating the benchmark to ensure that predictions are t'ruly out-of-sample and free from look-ahead bias. Specifically, a calendar of expected earnings announcements (EAs) is maintained to facilitate the collection of a fair ex ante information set, which is used to approximate Ω_{t-} prior to an event. Once an EA occurs, we collect the transcript from the event and provide it to candidate systems–either in its complete form or summarized.

We want to highlight that this strict temporal protocol applies at the observation level. Thus, information learned earlier in a given evaluation period can—*and likely should*—be used when making predictions later in the period. Along these lines, our initial evaluations use the first half of a quarter to optimize AI systems, which are then applied to the second half of the quarter. We find that this leads to a significant increase in performance. Given these results, while we choose to split the quarter evenly, there is no reason that this optimization and learning cannot be applied online in real-time.

3 Data and Results

3.1 Data Sources and Sample

Our initial implementation of the benchmark and results covers the fourth calendar quarter of 2025, i.e., earnings announcements from October through December. We choose to present results for this quarter because it is the first full quarter for which our production version of the benchmark was "frozen" and operational. This enables us to collect data, generate predictions, and evaluate results strictly out of sample.

The data for the benchmark are sourced from three real-time providers. First, we collect expected earnings events from Benzinga, which provides upcoming announcement dates, analyst consensus estimates $\mathbb{E}[e_{i,t}|\Omega^a_{t-}]$, and reported actuals, $e_{i,t}$. Conference call transcripts $\omega_{i,t}$—the primary textual input to the AI systems—come from S&P Capital IQ. Pricing data, including close-to-close returns for individual stocks and for the VTI market proxy, are obtained from Massive (formerly Polygon.io).

Outputs from AI systems are collected as soon as conference call transcripts become available. Specifically, following Section 2, we pass each transcript to the candidate AI systems and require predictions to be returned within a fixed time window. Because the benchmark is evaluated in real time, there is effectively no possibility that announcement returns or post-event commentary contaminate the information set provided to any model.[2]

[2]For Opus 4.5, we do process transcripts with a short delay relative to the other models. While this

The final sample comprises 1,849 earnings events spanning all 12 Fama–French industries and six reporting weeks within the quarter. Table 1 provides summary statistics for the sample. The median firm in our sample has a stock price of $29.66 and is covered by six analysts, while the mean coverage is 8.30 analysts. In robustness tests reported in the appendix, we verify that the choice of real-time data provider does not materially affect our results: substituting I/B/E/S for analyst estimates and actuals, and Capital IQ for pricing data, yields coefficient estimates and R^2 values that are not significantly different at the 1% level—although the sample size changes slightly due to differences in coverage.

3.2 Sample Characteristics

As a first step, we analyze the composition of our sample across industry and reporting time within the quarter (see Panel A of Table 1). Three industries account for more than half of the sample: Business Equipment (361 observations), Health (319), and Finance (300). Panel A also reveals substantial heterogeneity in the timing of announcements across industries. Financial firms report markedly earlier than the rest of the market, contributing 36 of the 49 observations in week 2 and 89 of the 222 in week 3. This early clustering implies that any model seeking to explain returns in the first few weeks of earnings season must primarily understand the financial sector. By contrast, Health firms are concentrated later in the quarter: 254 of their 319 announcements (80%) fall in weeks 5 and 6. Business Equipment firms follow a similar late-reporting pattern, with 198 of 361 events occurring in week 5 alone. The busiest week overall is week 5, which contains 850 observations—46% of the entire sample—reflecting the well-documented tendency for the majority of firms to cluster their announcements in the middle of the reporting window.

Panel B reports week-level summary statistics for abnormal returns ($r_{i,t}^{abn}$), earnings surprises ($s_{i,t}$), and the R^2 from regressing percentile-ranked abnormal returns on percentile-ranked surprises. Two patterns stand out. First, the earliest announcers deliver the most positive news. Week 1 firms have a mean abnormal return of 6% and a mean earnings surprise of 0.18% of price, both substantially above the full-sample averages of −1% and −0.05%, respectively. In contrast, the latest reporters deliver the most negative surprises: the mean surprise in week 6 is −0.58% of price. Part of these patterns may be consistent with the voluntary disclosure literature that predicts that managers with favorable news accelerate their announcements, while those with unfavorable results delay.

lag could in principle introduce look-ahead bias if the model's training corpus contained post-announcement information, Anthropic has confirmed that the training data cutoff for Opus 4.5 is August 2025, well before our sample period begins. Further, we do not allow Opus to search the web or access external sources of information. Therefore, the risk of leakage is quite small.

Second, the cross-sectional dispersion in earnings surprises increases dramatically across the quarter, rising from a standard deviation of 0.11 in week 1 to 6.59 in week 6. The R^2 from the surprise-return regression peaks at 0.16 in week 3—the week dominated by financial firms—and settles to 0.05 for the full sample. Given the heterogeneity across weeks, we do not place strong weight on the week-by-week variation in R^2; instead, the full-sample R^2 of 0.05 serves as our primary benchmark for evaluating AI models.

Panel C reports cross-sectional descriptive statistics for key firm characteristics. The typical firm in our sample is covered by six analysts (median), though coverage varies widely, with a standard deviation of 6.20 and an inter-quartile range of 3 to 12 analysts. Dispersion in analyst forecasts—a proxy for ex ante uncertainty about earnings—has a median of 0.04 and a mean of 0.09, indicating that a small number of firms with highly uncertain earnings pull the mean well above the median. Earnings surprises exhibit a similar pattern: the median surprise is positive at 0.12% of price, suggesting that the typical firm modestly beats consensus expectations, while the mean is −0.05%, dragged down by a left tail of large negative surprises concentrated among late-reporting firms.

3.3 Initial Results

Our results from the first iteration of the benchmark are focused on evaluating three frontier LLMs: Google's Gemini 2.5 Flash-Lite, OpenAI's GPT-5 nano, and Anthropic's Opus 4.5. We begin by assessing their baseline ability to explain contemporaneous stock returns around earnings announcements, i.e., how good are the models when given comparatively simple prompts. We then examine the effects of transcript summarization, event timing, and industry heterogeneity. Throughout, we report R^2 from regressions of abnormal returns on model predictions and the earnings surprise, with 99% bootstrap confidence intervals in brackets.

3.3.1 LLM Predictions Using Full Transcripts

Table 2 presents our baseline results using full earnings call transcripts. Column (1) establishes the benchmark: regressing percentile-ranked abnormal returns on the percentile-ranked earnings surprise alone yields an R^2 of 5.0%, consistent with the upper end of estimates in the existing literature. The remaining columns evaluate the three LLMs, first in isolation and then jointly with the surprise.

Even without any optimization, LLM predictions substantially increase explanatory power. Opus 4.5 achieves the highest standalone R^2 of 10.9% [0.08, 0.15], more than doubling the surprise-only baseline. Gemini 2.5 Flash-Lite and GPT-5 nano perform similarly to each

other, with standalone R^2 values of 7.5% and 7.1%, respectively. Notably, GPT-5 nano achieves its R^2 with a much larger coefficient on its prediction (0.726 versus 0.374 for Gemini), suggesting that the two models use different scales but extract comparable amounts of cross-sectional information from the transcript.

A key finding is that LLM predictions and the earnings surprise are complementary. When we include both the LLM prediction and the surprise in the regression (columns 5–7), the coefficient on the surprise remains large and highly significant for all three models, declining only modestly from 0.219 to approximately 0.18–0.19. This indicates that the models capture information in the transcript that goes beyond simple earnings news—likely including guidance, management tone, and the question-and-answer discussion—while the quantitative surprise retains independent explanatory power. The combined specification using Opus 4.5 yields an R^2 of 14.3% [0.10, 0.18], nearly tripling the surprise-only benchmark.

Figure 1 places these results in historical context. The literature has progressed from an R^2 of roughly 2% in Lev (1989) to approximately 4% in Chiang et al. (2019) and 7.5% in Hand et al. (2022), where the latter result required incorporating surprises along 33 dimensions beyond earnings. Our best unoptimized model achieves 14.3% using only the earnings surprise and a single LLM prediction derived from the transcript. This represents a significant advance in the ability to explain the "dark matter" of stock price movements around earnings announcements. More broadly, these initial findings already highlight that the benchmark that we propose in this paper can be improved using modern AI systems, yet leaving scope for improvement. This makes it an attractive target for future research.

3.3.2 Summarized Transcripts

Table 3 repeats the analysis using transcript summaries generated by Opus 4.5 in place of full transcripts. These summaries are approximately 28 times shorter than the original documents, which has important implications for both cost and the feasibility of prompt optimization, as optimization methods require many evaluations of the objective function.

The results show that summaries preserve most of the information relevant for explaining returns. Opus 4.5 applied to its own summaries achieves a standalone R^2 of 9.3% [0.06, 0.13], compared to 10.9% with the full transcript. When combined with the surprise, the summary-based specification yields an R^2 of 13.0% [0.09, 0.18] versus 14.3% with the full transcript—a decline of only 1.3 percentage points, or roughly 9% in relative terms. The bootstrap confidence intervals for the two specifications overlap substantially, suggesting that the difference is not statistically significant at the 1% level.

The loss from summarization is more pronounced for the other two models. GPT-5 nano's standalone R^2 drops from 7.1% to 3.0% when using summaries, indicating that this

model is more reliant on details present in the full transcript that are lost during summarization. Gemini 2.5 Flash-Lite exhibits a more moderate decline, from 7.5% to 6.5%. These differences across models suggest that the quality of the summarization step interacts with the downstream prediction model, a question we leave for future investigation. For our optimization experiments below, we proceed with summaries given their substantially lower cost and the modest loss in explanatory power for the best-performing model.

3.3.3 Early versus Late Announcements

Our optimization methods require a training set to learn from and a separate evaluation set to test on. We implement this by splitting the quarter at the median announcement date, yielding an "Early" subsample of 1,032 observations and a "Late" subsample of 817 observations. The early period is used for training, and the late period serves as the out-of-sample evaluation set.

Table 5 reports results from this split. A natural concern is that model performance might differ systematically across the two subperiods, which would complicate interpretation of the optimization results. Reassuringly, we find that performance is similar or modestly better in the late period. For Opus 4.5 with summaries and surprise, the R^2 is 11.8% in the early period and 14.1% in the late period. Gemini 2.5 Flash-Lite shows a comparable pattern (9.3% early, 10.5% late), as does GPT-5 nano (6.6% early, 8.5% late). The slightly higher R^2 in the late period may reflect the compositional shift toward industries where LLMs perform particularly well, or simply sampling variation. The key takeaway is that there is no evidence that the evaluation period is systematically harder than the training period, which would bias against finding optimization gains.

3.4 Optimization

The results presented thus far are based on comparatively simple and basic prompts, e.g., "Your task is X, be mindful of Y and Z, the typical steps to performing X are 1, 2, and 3. Now, go do it." Recent work in the computer science literature has shown that significant increases in performance can be achieved through minor changes to the prompts, e.g., encouraging the model to think before answering (Wei et al., 2022), or gradient-free optimization of prompts, e.g., Genetic-Pareto Prompt Optimization (Agrawal et al., 2026) and Agentic Context Engineering (ACE) (Zhang et al., 2026). In this section, we explore the value of model context optimization within the setting of the benchmark.

3.4.1 Setup

Given that performance on the benchmark is measured by the R^2 of the regression defined in Eq. (3), our primary objective is to maximize this out-of-sample explanatory power by varying the context prompt p provided to the LLM. To ensure rigorous evaluation and to prevent overfitting, we partition our sample of discrete information events into a training set, $\mathcal{E}_{train}$, and a validation set, $\mathcal{E}_{val}$. We index individual events (unique firm-time observations) by k.

To make prompt optimization computationally tractable, we adopt a two-stage approach on the training data. First, we estimate the baseline regression model strictly over the training set:

$$\epsilon_k = \alpha + \boldsymbol{\theta}^\top \mathbf{X}_k + u_k, \quad \forall k \in \mathcal{E}_{train}. \tag{9}$$

We then extract the residuals, u_k, which represent the portion of the abnormal returns unexplained by the controls. In the second stage, we optimize the prompt p by minimizing the mean squared error (MSE) between the AI system's predictions and these residuals. Formally, we search for the optimal prompt p^*:

$$p^* = \arg\min_{p \in \mathcal{P}} \frac{1}{|\mathcal{E}_{train}|} \sum_{k \in \mathcal{E}_{train}} (u_k - \hat{g}_p(\omega_k, \Omega_k))^2, \tag{10}$$

where $\mathcal{P}$ is the set of all possible prompts, and $\hat{g}_p(\cdot)$ is the scalar prediction from the LLM given prompt p.

After identifying p^*, we generate out-of-sample predictions, $\hat{g}_{p^*}(\omega_k, \Omega_k)$, for all events in the validation set. The final benchmark performance is then evaluated by estimating the full cross-sectional regression on $\mathcal{E}_{val}$:

$$\epsilon_k = \alpha_{val} + \gamma \hat{g}_{p^*}(\omega_k, \Omega_k) + \boldsymbol{\theta}_{val}^\top \mathbf{X}_k + \nu_k, \quad \forall k \in \mathcal{E}_{val}, \tag{11}$$

and calculating the R^2. We split the data from calendar quarter four of 2025 into training and validation sets, i.e., $\mathcal{E}_{train}$ and $\mathcal{E}_{val}$, based on the median date following Section 3.3.3.

While conceptually simple, there are several practical challenges to implementing solutions to Eq. (10). One of the most notable is the lack of continuous gradient information to steer changes to the prompt. As such, we focus our attention on exploring two state-of-the-art, gradient-free methods of prompt optimization: Genetic-Pareto (GEPA) search (Agrawal et al., 2026) and Agentic Context Engineering (ACE) (Zhang et al., 2026).

3.4.2 Genetic-Pareto Prompt Optimization

GEPA (Agrawal et al., 2026) is a prompt optimization algorithm that iteratively improves the instructions given to an LLM, i.e., the prompt or context p above. The method maintains a population of candidate prompts and evolves them through repeated selection, mutation, and evaluation. At each iteration, GEPA selects a candidate from the Pareto frontier—the subset of prompts that each achieve the highest score on at least one training instance—sampling in proportion to the number of instances on which each candidate leads. This selection mechanism encourages diversity across the candidate pool.

Once a prompt has been selected, it is applied to a mini-batch of training examples, and a scoring function identifies where the model succeeds and fails, returning textual diagnostics describing failure patterns. An LLM then reads the current prompt alongside these diagnostics and proposes a targeted revision, i.e., "reflective mutation." This reflection step involves the model itself diagnosing shortcomings in natural language rather than relying on random perturbations or gradient signals. If the revised prompt improves performance on the mini-batch, it enters the candidate pool; otherwise it is discarded. The algorithm terminates when a computational budget is exhausted and returns the candidate with the highest average score on a held-out evaluation set.

Table 6 reports results from applying Genetic-Pareto (GEPA) prompt optimization to the training dataset $\mathcal{E}_{train}$ and evaluating Gemini 2.5 Flash-Lite with the optimized prompt out-of-sample on $\mathcal{E}_{val}$. During training, GEPA achieves a roughly 30% reduction in the MSE objective, indicating that the optimization procedure does meaningfully alter the model's predictions. However, the R^2 improvement in the evaluation set is modest. The baseline specification (Gemini with the default prompt, plus surprise) yields an R^2 of 10.5%, while the GEPA-optimized specifications achieve 12.8% under both loss functions. The 99% bootstrap confidence intervals for the baseline [0.06, 0.16] and optimized [0.08, 0.19] specifications overlap, although the point estimates suggest a gain of approximately 2.3 percentage points.

The reason for the gap between training improvement and evaluation R^2 becomes apparent upon decomposing the MSE objective and examining the optimized prompt. Beginning with the former, the MSE objective can be written as:

$$MSE(u, \hat{g}) = Var(u) + Var(\hat{g}) + 2Cov(u, \hat{g}) + \mathbb{E}[\hat{g}^2]. \tag{12}$$

As such, meaningful decreases in the objective can come from improving the covariance between the predictions and the residual—which would improve the R^2—or through other channels such as de-biasing the model, i.e., making $\mathbb{E}[\hat{g}] = 0$. Our empirical results suggest that de-biasing is the "path of least resistance" in our setting.

We can also examine the optimized prompt (see Figure 2) to explore whether the primary effect of GEPA is to de-bias the model's predictions. Doing so reveals the core directive which emerges from the optimization: "good is priced in." Effectively, the model learns that its default predictions systematically overestimate the market's response to strong operational results.

Specifically, the optimized prompt instructs the model to treat companies that report record profits and 20% growth as delivering a neutral outcome, not a positive one, because such performance is already reflected in the stock price. The prompt further develops sector-specific heuristics: for telecom and utility stocks, it caps predictions at 0.50 when subscriber churn rises, even if net additions are strong; for high-growth technology firms, it discounts long-term product roadmap promises in favor of immediate financial delivery; and for financials, it treats share buybacks as standard capital maintenance rather than positive catalysts.

Given the observed tendency of the model to focus on de-biasing, we additionally explore an optimization objective where we subtract the covariance between the residual and the model's predictions. This has the effect of increasing the weight placed on the $Cov(u, \hat{g})$ term in Eq. (12). We do not find a significant increase in the R^2 under this objective, suggesting that directly optimizing for increased R^2 may be necessary—although debiasing alone is a valuable result.

These findings illustrate both a strength and a limitation of prompt optimization. The strength is interpretability: the optimized prompt reveals systematic biases in the model's default reasoning about markets. The limitation is that correcting a mean bias, while valuable, contributes less to R^2 than improving the model's ability to rank outcomes correctly. De-biasing primarily shifts the intercept of the prediction, whereas R^2 is driven by the correlation between predicted and actual returns.

3.4.3 Agentic Context Engineering (ACE)

ACE, introduced by Zhang et al. (2026), is a framework for optimizing the context provided to an LLM by building a structured collection of rules—termed a "playbook" or "rulebook"—through an iterative generate-reflect-curate loop. In each iteration, a Generator module receives the current playbook and a task instance, attempts the task while recording its reasoning, and flags which existing rules proved useful or misleading. A Reflector module then analyzes this record, identifies patterns in errors and successes, and proposes targeted additions or modifications—small "delta" updates rather than a wholesale rewrite of the context. Finally, a Curator module integrates these deltas into the playbook: new rules are appended, existing rules are updated based on whether they improved or degraded performance, and

duplicate or overlapping rules are removed.

This incremental architecture addresses a key failure mode of monolithic prompt optimization: because each iteration modifies only a small portion of the context, the method avoids "context collapse," in which successive LLM rewrites progressively compress and discard accumulated knowledge. The result is a modular, interpretable set of rules that grows and refines over time.

In Table 7, we find that ACE delivers a material improvement in explanatory power. Without ACE, Opus 4.5 with the surprise achieves an R^2 of 14.1% [0.09, 0.20] on the late subsample. With the ACE rulebook, the R^2 rises to 17.1% [0.12, 0.24], a gain of 3.0 percentage points. The coefficient on the LLM prediction is essentially unchanged (0.481 versus 0.484), while the coefficient on the surprise declines slightly from 0.175 to 0.166, suggesting that ACE enables the model to capture some information that was previously reflected only in the quantitative surprise. The R^2 of 17.1% represents more than a threefold improvement over the surprise-only baseline and, to our knowledge, the highest explanatory power reported in the earnings announcement return literature.

The ACE rulebook, like the GEPA prompt, is interpretable. It contains specific, actionable heuristics that reflect how markets actually respond to different types of earnings news (see Figure 3). For instance, the rulebook instructs the model that when a pre-profit company reports revenue growth exceeding 30% year-over-year but operating losses remain flat, the prediction should fall in the 0.10–0.20 range, because markets demand that high growth translate into financial improvement. Similarly, when management uses cautionary forward-looking language—phrases such as "aggressive consensus" or "gradual recovery"—the rulebook directs predictions to the 0.35–0.50 range. The rulebook also identifies interaction effects: when record margins are attributed to non-recurring benefits (such as precious metals revaluations) and the same quarter features an operational execution failure alongside a major geographic segment declining by more than 15%, the prediction should be 0.10–0.20 regardless of positive demand narratives elsewhere in the transcript. These rules echo well-established findings in the accounting and finance literature regarding the market's differential response to transitory versus permanent earnings components, but they are discovered entirely from data by the agentic system.

3.4.4 Model Size and Reasoning Effort

A natural question is whether the gains from ACE depend on the size and capability of the model that applies the rulebook. Table 8 addresses this by evaluating the ACE rulebook—developed using Opus 4.5 at high reasoning effort—with smaller models (Haiku 4.5 and Sonnet 4.5) and with Opus 4.5 at reduced reasoning effort.

The results reveal an important pattern: the ACE rulebook effectively transfers knowledge from a strong model to weaker ones. Without ACE, Opus 4.5 at high reasoning effort achieves an R^2 of 14.1%. With the ACE rulebook, Haiku 4.5—a substantially smaller and cheaper model—achieves 11.7%, Sonnet 4.5 achieves 12.1%, and Opus 4.5 at low reasoning effort achieves 13.9%. The last result is particularly striking: Opus 4.5 at low effort with ACE (13.9%) nearly matches Opus 4.5 at high effort without ACE (14.1%), despite requiring far less computation at inference time. This suggests that much of the value added by the most capable model lies in the structured reasoning that the ACE rulebook captures explicitly, rather than in the raw scale of the model's parameters or the depth of its inference-time computation.

These findings have practical implications for deploying AI systems to analyze earnings announcements at scale. The ACE rulebook can be developed once per quarter using the most capable model during the training period, and then applied to a cheaper, faster model for the remainder of the quarter. This "distillation through context" approach offers a favorable cost-performance tradeoff: the smaller models with ACE achieve R^2 values of 11.7–13.9%, compared to 14.1% for the full-scale model without ACE, at a fraction of the inference cost.

4 On the Differences between Agentic Optimization and Traditional Machine Learning

The agentic optimization methods that we explore provide, at least, two important benefits over traditional machine learning (ML) methods. The first benefit relates to efficiency. Both methods leverage contextualized feedback during optimization, i.e., after a model reasons about a specific instance of a forecasting problem and makes a prediction, its prediction is compared to the realized value and feedback is given on why the prediction was good or bad. This feedback—contextualized by the specific instance at hand—allows an LLM to learn more efficiently from even a single observation.[3]

To see this, consider a binary classification task where traditional ML methods learn by minimizing some loss function over predictions and targets: 0's and 1's (e.g., successful and unsuccessful predictions). The only feedback in this setting is the gradient of the loss, which for any single observation may be incredibly noisy. In contrast, ACE and GEPA explicitly reflect on *why* the prediction was (un)successful, a process which leverages the

[3] Agrawal et al. (2026) illustrate the efficiency gains of GEPA compared to traditional reinforcement learning methods.

existing knowledge in the LLM and any information encoded in the current version of the rulebook or prompt.

To provide a simple analogy, a child learning math who receives only "correct" or "incorrect" as feedback, may take a long time to master the subject. If, instead, the child receives additional feedback *why* a particular answer is correct or not, she will learn more efficiently. In our context, an LLM reflecting on why a particular forecast is too optimistic or pessimistic may lead to rules that generalize to other earnings calls.

AI-based optimization provides a related benefit by being able to leverage knowledge acquired during pretraining regarding the economic meaning of features and how they are related. In contrast, traditional ML methods are unaware of the relationship between variables and those need to be learned from data. Thus, LLMs' basic understanding how certain variables are related may help in generating predictions. As a simple example, if we provide an agentic AI system information on sales and cost of goods sold, it will understand that it also has information about profitability and margins. A machine learning algorithm does not have this prior economic understanding and context, and will have to learn it from data.

The second key benefit is that the optimized prompts and context may identify new determinants of realized returns around key news events. In particular, asset pricing models have implications for realized returns, in addition to implications for moments of the return distribution such as expected returns and (co)variances. If the rulebooks learned by ACE or the prompts learned by GEPA do not have natural counterparts in current asset pricing models, it may be worth considering such extensions. Further, the optimized context often provides rules that are specific to a certain industry, which may be helpful to develop industry-specific models. As a concrete example, if changes in guidance are of first-order importance to explain announcement returns, asset pricing models ideally account for guidance provided by firms to investors. This fits in a rich tradition to explore models of beliefs for investors and firm management, potentially going beyond rational expectations.

As the prompts and context are learned from data, AI-based optimization adds a valuable tool to the toolbox of empirical asset pricing researchers by integrating empirical learning with economic structure.

5 A New Competition to Explain Asset Prices

Our results thus far suggest that AI systems can materially increase our understanding of asset prices around earnings announcements. While these results are a contribution on their own, we do not intend for them to be the "definitive" results. For example, we anticipate further progress from including historical events, incorporating peers, or improved finance-

specific harnesses for agents.

Rather than working on these in a silo, we are introducing a competition with the explicit goal of improving our understanding of asset prices around information releases—related, Hellum et al. (2025) introduce a broad competition to identify the pricing kernel, in contrast, our aim is focused on understanding how information is priced. This follows a broader tradition in asset pricing to benchmark asset pricing models by their ability to explain differences in average returns. However, given the volatility of returns, it is challenging to discriminate between models during a short out-of-sample period. Our benchmark instead can be evaluated during a single earnings season, which allows for the rapid evolution of optimized AI systems.[4]

In line with the principles in Section 2, the competition is open to anyone—the cost of running Gemini in our experiments was much less than $10 for an entire earnings season—and evaluated transparently. Participants need only register their system with the competition on explainingmarkets.ai to receive programmatic access to (i) a common EA calendar and (ii) an opportunity to log a prediction in real-time when an EA occurs. By defining a common set of events, evaluation methodology, and introducing a competition, we aim to spur progress on our understanding of asset prices similarly to how the introduction of the ImageNet Large Scale Visual Recognition Challenge (Deng et al., 2009) (i) led to dramatic decreases in the error rates of supervised models over only a few years (Russakovsky et al., 2015), and (ii) supported subsequent development of scalable self-supervised learning methods within computer vision (see, e.g., Chen et al., 2020; Caron et al., 2021).

6 Conclusions

Evaluating optimized AI systems for asset pricing is fundamentally difficult for at least two reasons. First, AI models are trained on all data, implying that any backtest or analysis using historical data suffers from look-ahead bias. In addition, markets are reflexive — as investors adopt AI, prices adjust — which may erode the very patterns the AI system was trained to exploit. We introduce a real-time, out-of-sample benchmark designed to sidestep both problems. The benchmark measures how well AI systems can explain contemporaneous stock returns around earnings announcements using only information available at announcement time, including the announcement text itself.

Applying this benchmark to a range of agentic AI systems — which extract signals from earnings call transcripts and optimize over those signals — we find that the best-optimized

[4]See also Gabaix et al. (2025) for cross-sectional benchmarks of models of asset and investor embeddings, which are methods to represent assets and investors beyond traditional characteristics.

systems more than double the explained variation in returns relative to standard benchmarks (R^2 rising from 8% to close to 20%). Applying this benchmark to a range of agentic AI systems — which extract signals from earnings call transcripts and optimize over those signals — we find that the best-optimized systems more than double the explained variation in returns relative to standard benchmarks (R^2 rising from 8% to close to 20%). Further, we show that AI-based optimization can deliver efficiency gains relative to traditional machine learning methods while also improving interpretability as our approach produces human-readable economic mechanisms that explain price movements. These learned rules can be compared to the drivers of realized returns in existing asset pricing models to potentially identify missing sources of variation in a data-driven, self-evolving way that integrates empirical learning with economic structure. We release an SDK for researchers to improve on our results. Saturating this benchmark would represent fundamental progress in understanding how capital markets process firm-level information around key news events.

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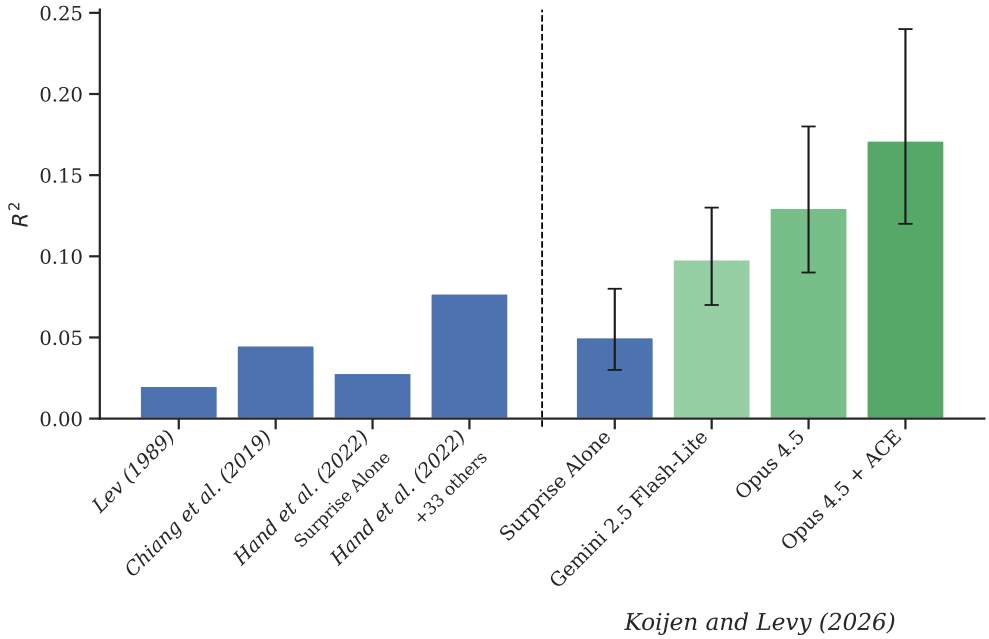
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Figure 1: Progress on Explaining Asset Prices Around Earnings Announcements



Notes: For each paper, this figure plots the R^2 from a regression of abnormal returns over a narrow window around the earnings announcement on measures found to explain the returns.

Figure 2: Excerpt from the GEPA-Optimized Prompt

This figure presents excerpts from the prompt produced by Genetic-Pareto (GEPA) optimization, trained on $\mathcal{E}_{train}$ using Gemini 2.5 Flash-Lite as the target model. Panel A shows the core directive discovered by the optimization. Panel B presents sector-specific heuristics derived from past prediction errors. See Table 6 for evaluation results. Carriage returns indicated by "↪" are solely to indicate line wrapping in this presentation and are not part of the prompt.

Panel A: Core directive

The Core Directive: Assume "Good" is "Priced In"

Your previous predictions have consistently failed by overestimating the market's excitement for strong operational results.

* The Baseline is Success: Public companies are expected to grow, beat estimates, and improve margins. A company reporting record profits and 20% growth is usually just meeting the high expectations already built into the stock price. This is a Neutral (0.40 – 0.60) result, not a positive one.
* Neutral = The Graveyard of Good News: If a company reports a "beat and raise" that is merely incremental, or if they report strong earnings but flat guidance, the stock will likely not move or may even drift lower as traders "sell the news."
* The Bar for "Up" is Extremely High: To justify a prediction > 0.75, the news must be shocking (e.g., doubling guidance, a massive surprise buyback >10% of market cap, or a breakthrough product launch today).

Panel B: Sector-specific heuristics

Sector–Specific Heuristics (based on past errors):

1. Telecom/Utilities (e.g., AT&T): These are defensive yield stocks. High subscriber growth (net adds) is often ignored if churn ticks up even slightly or if free cash flow guidance isn't raised. The market fears competition more than it rewards growth here. Heuristic: If churn rises, cap prediction at 0.50.
2. High–Growth/Tech (e.g., Tesla): These stocks have massive premiums. Operational improvements (cash flow, margins) are expected. "Roadmap" items (future products years away) are often discounted as "vaporware" in the short term. Heuristic: Discount long–term promises. Only value immediate financial delivery.
3. Financials/Insurance (e.g., Travelers): Buybacks are standard operating procedure, not surprises. A $3B buyback is often just capital maintenance. "Underwriting discipline" (shrinking volume to maintain price) is often viewed as a growth headwind. Heuristic: Without a massive dividend hike or distinct catalyst, stick to 0.40–0.50.

Figure 3: Example Rules from the ACE Rulebook

This figure presents three representative rules from the rulebook produced by Agentic Context Engineering (ACE), trained on $\mathcal{E}_{train}$ using Opus 4.5 as the target model. Each rule specifies conditions observed in earnings call transcripts and the prediction range the model should assign. See Table 7 for evaluation results. Carriage returns indicated by "↪" are solely to indicate line wrapping in this presentation and are not part of the rulebook.

* When a pre−profit company reports strong revenue growth (>30% YoY) BUT operating
  ↪ losses remain flat or worsen (EBITDA/operating loss not narrowing proportionally),
  ↪ apply 0.10−0.20 range. Markets demand that high growth translate to financial
  ↪ improvement. Amplified when: (1) forward guidance suggests growth momentum
  ↪ peaking (flat sequential); (2) in hot theme sector (LiDAR, autonomy, quantum, etc.);
  ↪ (3) customer conversion rates below 15%.

* When management uses cautionary forward language ("aggressive consensus," "gradual
  ↪ recovery," guiding to low end), apply 0.35−0.50. When forward quarter guidance
  ↪ represents >50% deceleration from recent organic growth, apply 0.10−0.25. When
  ↪ one−time items inflate headline EPS with no forward acceleration, apply 0.10−0.20.
  ↪ CRITICAL: premium brand guidance below framework + DTC decline = 0.05−0.15.

* When record or near−record margins include acknowledged non−recurring benefits (precious
  ↪ metals, inventory revaluations, one−time mix) AND there is a concrete operational
  ↪ execution failure in the same quarter (equipment downtime, supply disruption, facility
  ↪ issue) AND a major geographic segment is declining >15% from external headwinds,
  ↪ apply 0.10−0.20 range regardless of multi−sector demand narratives.

Table 1: Summary Statistics for Q4 2025 Earnings Announcements. Panel A reports the number of unique earnings events by reporting week and Fama-French 12 industry classification. Weeks are indexed starting from the first week with observations in the quarter. Panel B reports the weekly mean, median, and standard deviation of one-day cumulative abnormal returns ($r_{i,t}^{\text{abn}}$) and the earnings surprise expressed as a percentage of price ($s_{i,t}$), along with the R^2 from a regression of percentile-ranked abnormal returns on percentile-ranked surprise. Panel C reports cross-sectional descriptive statistics for key firm characteristics. All continuous variables are winsorized at the 1st and 99th percentiles.

Panel A: Number of Observations by Week and Industry

Week	NoDur	Durbl	Manuf	Enrgy	Chems	BusEq	Telcm	Utils	Shops	Hlth	Fin	Other	Total
1	2	0	2	0	0	0	0	0	0	1	0	1	6
2	0	1	4	1	1	1	0	0	1	1	36	3	49
3	8	6	26	9	6	23	3	4	10	10	89	28	222
4	16	16	62	18	12	80	7	19	39	53	77	59	458
5	35	13	57	30	30	198	17	32	57	169	82	130	850
6	6	7	21	7	3	59	6	2	16	85	16	36	264
Total	67	43	172	65	52	361	33	57	123	319	300	257	1849

Panel B: Weekly Statistics

		$r_{i,t}^{\text{abn}}$			$s_{i,t}$ (%)			
Week	N	Mean	Med.	SD	Mean	Med.	SD	R^2
1	6	0.06	0.05	0.12	0.18	0.17	0.11	
2	49	-0.00	-0.00	0.07	0.13	0.12	0.16	0.00
3	222	0.00	-0.01	0.08	0.13	0.10	1.26	0.16
4	458	-0.01	-0.01	0.10	0.11	0.10	1.31	0.05
5	850	-0.01	-0.01	0.12	-0.02	0.14	2.93	0.05
6	264	-0.01	-0.02	0.12	-0.58	0.15	6.59	0.03
Total	1,849	-0.01	-0.01	0.11	-0.05	0.12	3.28	0.05

Panel C: Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75
Price	1,849	71.41	112.68	8.67	29.66	79.98
$r_{i,t}^{\text{abn}}$	1,849	-0.01	0.11	-0.06	-0.01	0.05
Coverage	1,796	8.30	6.20	3.00	6.00	12.00
Dispersion	1,694	0.09	0.16	0.02	0.04	0.10
$s_{i,t}$	1,849	-0.05	3.28	-0.03	0.12	0.44

Table 2: Explaining Contemporaneous Stock Returns Using Full Transcripts

	Dependent variable: $r_{i,t}^{abn}$						
	Baseline	Gemini 2.5 Flash-Lite	GPT-5 nano	Opus 4.5	Gemini 2.5 Flash-Lite	GPT-5 nano	Opus 4.5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
α	0.384***	0.206***	0.026	0.140***	0.135***	-0.023	0.075***
	(0.013)	(0.024)	(0.040)	(0.024)	(0.025)	(0.039)	(0.025)
$s_{i,t}$	0.219***				0.190***	0.187***	0.181***
	(0.022)				(0.022)	(0.022)	(0.021)
$\hat{g}(\omega_{i,t}, \Omega_{t-})$		0.374***	0.726***	0.608***	0.343***	0.656***	0.565***
		(0.031)	(0.061)	(0.040)	(0.030)	(0.060)	(0.040)
R^2	0.050	0.075	0.071	0.109	0.112	0.107	0.143
	[0.02, 0.08]	[0.05, 0.11]	[0.04, 0.10]	[0.08, 0.15]	[0.08, 0.15]	[0.08, 0.15]	[0.10, 0.18]
Adjusted R^2	0.050	0.074	0.071	0.109	0.111	0.106	0.142
	[0.02, 0.08]	[0.05, 0.11]	[0.04, 0.10]	[0.08, 0.15]	[0.08, 0.15]	[0.08, 0.15]	[0.10, 0.18]
Observations	1849	1849	1849	1849	1849	1849	1849
F Statistic	97.282***	149.676***	142.150***	226.852***	116.796***	111.038***	154.169***

Notes: This table shows the results of regressing abnormal returns ($r_{i,t}^{abn}$) on the earnings surprise ($s_{i,t}$) and various LLM-derived predictions ($\hat{g}(\omega_{i,t}, \Omega_{t-})$) of the $r_{i,t}^{abn}$ using the full earnings call transcript ($\omega_{i,t}$). Standard errors are presented in parentheses under coefficient estimates. 99% confidence intervals based on 1,000 bootstrap samples are presented in brackets under estimates.

Table 3: Transcript Summaries

	Dependent variable: $r_{i,t}^{abn}$						
	Baseline	Gemini 2.5 Flash-Lite	GPT-5 nano	Opus 4.5	Gemini 2.5 Flash-Lite	GPT-5 nano	Opus 4.5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
α	0.384***	0.283***	0.271***	0.210***	0.220***	0.198***	0.134***
	(0.013)	(0.020)	(0.030)	(0.022)	(0.021)	(0.030)	(0.023)
$s_{i,t}$	0.219***				0.180***	0.203***	0.189***
	(0.022)				(0.022)	(0.022)	(0.021)
$\hat{g}(\omega_{i,t}, \Omega_{t-})$		0.289***	0.295***	0.474***	0.252***	0.257***	0.442***
		(0.025)	(0.039)	(0.034)	(0.025)	(0.038)	(0.034)
R^2	0.050	0.065	0.030	0.093	0.098	0.073	0.130
	[0.03, 0.08]	[0.04, 0.10]	[0.01, 0.05]	[0.06, 0.13]	[0.07, 0.13]	[0.04, 0.11]	[0.09, 0.18]
Adjusted R^2	0.050	0.064	0.029	0.092	0.097	0.072	0.129
	[0.03, 0.08]	[0.04, 0.10]	[0.01, 0.05]	[0.06, 0.13]	[0.06, 0.13]	[0.04, 0.10]	[0.09, 0.17]
Observations	1849	1849	1849	1849	1849	1849	1849
F Statistic	97.282***	128.398***	57.137***	188.621***	100.273***	72.244***	137.546***

Notes: This table shows the results of regressing abnormal returns ($r_{i,t}^{abn}$) on the earnings surprise ($s_{i,t}$) and various LLM-derived predictions ($\hat{g}(\hat{\omega}_{i,t}, \Omega_{t-})$) of the $r_{i,t}^{abn}$ using summaries of the earnings call transcript ($\hat{\omega}_{i,t}$). Summaries are created using each respective model. Standard errors are presented in parentheses under coefficient estimates. 99% confidence intervals based on 1,000 bootstrap samples are presented in brackets under estimates.

Table 4: Opus 4.5 Transcript Summaries

	Dependent variable: $r_{i,t}^{abn}$						
	Baseline	Gemini 2.5 Flash-Lite	GPT-5 nano	Opus 4.5	Gemini 2.5 Flash-Lite	GPT-5 nano	Opus 4.5
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
α	0.384***	0.288***	0.229***	0.210***	0.211***	0.154***	0.134***
	(0.013)	(0.018)	(0.024)	(0.022)	(0.020)	(0.025)	(0.023)
$s_{i,t}$	0.219***				0.189***	0.194***	0.189***
	(0.022)				(0.022)	(0.022)	(0.021)
$\hat{g}(\omega_{i,t}, \Omega_{t-})$		0.289***	0.358***	0.474***	0.264***	0.329***	0.442***
		(0.024)	(0.031)	(0.034)	(0.023)	(0.031)	(0.034)
R^2	0.050	0.075	0.067	0.093	0.112	0.106	0.130
	[0.03, 0.08]	[0.05, 0.11]	[0.04, 0.10]	[0.06, 0.13]	[0.08, 0.15]	[0.07, 0.15]	[0.09, 0.18]
Adjusted R^2	0.050	0.075	0.067	0.092	0.111	0.105	0.129
	[0.03, 0.08]	[0.05, 0.11]	[0.04, 0.10]	[0.06, 0.13]	[0.08, 0.15]	[0.07, 0.15]	[0.09, 0.17]
Observations	1849	1849	1849	1849	1849	1849	1849
F Statistic	97.282***	150.408***	132.875***	188.621***	116.445***	109.305***	137.546***

Notes: This table shows the results of regressing abnormal returns ($r_{i,t}^{abn}$) on the earnings surprise ($s_{i,t}$) and various LLM-derived predictions ($\hat{g}(\hat{\omega}_{i,t}, \Omega_{t-})$) of the $r_{i,t}^{abn}$ using summaries of the earnings call transcript ($\hat{\omega}_{i,t}$). Summaries are created using Opus 4.5. Standard errors are presented in parentheses under coefficient estimates. 99% confidence intervals based on 1,000 bootstrap samples are presented in brackets under estimates.

Table 5: Announcement Timing: Early versus Late

	Dependent variable: $r_{i,t}^{abn}$					
	Gemini 2.5 Flash-Lite		GPT-5 nano		Opus 4.5	
	(1) Early	(2) Late	(3) Early	(4) Late	(5) Early	(6) Late
α	0.230***	0.207***	0.251***	0.138***	0.150***	0.118***
	(0.029)	(0.030)	(0.041)	(0.046)	(0.032)	(0.033)
$\hat{g}(\omega_{i,t}, \Omega_{t-})$	0.222***	0.276***	0.170***	0.345***	0.398***	0.481***
	(0.034)	(0.038)	(0.050)	(0.059)	(0.047)	(0.051)
$s_{i,t}$	0.218***	0.156***	0.244***	0.176***	0.213***	0.175***
	(0.033)	(0.030)	(0.033)	(0.030)	(0.032)	(0.029)
R^2	0.093	0.105	0.066	0.085	0.118	0.141
	[0.06, 0.14]	[0.06, 0.16]	[0.04, 0.11]	[0.05, 0.14]	[0.07, 0.18]	[0.09, 0.20]
Adjusted R^2	0.091	0.103	0.064	0.083	0.116	0.139
	[0.05, 0.14]	[0.06, 0.16]	[0.03, 0.11]	[0.04, 0.14]	[0.07, 0.17]	[0.09, 0.20]
Observations	1032	817	1032	817	1032	817
F Statistic	52.506***	47.772***	36.262***	38.040***	68.869***	66.897***

Notes: This table shows the results of regressing abnormal returns ($r_{i,t}^{abn}$) on the earnings ($s_{i,t}$) and various LLM-derived predictions ($\hat{g}(\omega_{i,t}, \Omega_{t-})$) of the $r_{i,t}^{abn}$ using summaries of the earnings call transcript ($\hat{\omega}_{i,t}$). Summaries are created using Opus 4.5. The sample from Table 3 is split based on the median date into "Early" and "Late" sub-samples. Standard errors are presented in parentheses under coefficient estimates. 99% confidence intervals based on 1,000 bootstrap samples are presented in brackets under estimates.

Table 6: Genetic-Pareto Prompt Optimization

	Dependent variable: $r_{i,t}^{abn}$					
	Baseline		Loss = MSE		Loss = MSE - Cov	
	(1)	(2)	(3)	(4)	(5)	(6)
α	0.261***	0.207***	0.296***	0.217***	0.223***	0.151***
	(0.029)	(0.030)	(0.023)	(0.026)	(0.030)	(0.032)
$\hat{g}_{p^*}(\omega_{i,t}, \Omega_{t-})$	0.310***	0.276***	0.369***	0.352***	0.443***	0.419***
	(0.038)	(0.038)	(0.041)	(0.040)	(0.049)	(0.048)
$s_{i,t}$		0.156***		0.175***		0.172***
		(0.030)		(0.029)		(0.029)
R^2	0.075	0.105	0.090	0.128	0.091	0.128
	[0.04, 0.13]	[0.06, 0.16]	[0.05, 0.15]	[0.08, 0.19]	[0.05, 0.14]	[0.08, 0.18]
Adjusted R^2	0.074	0.103	0.089	0.126	0.090	0.126
	[0.04, 0.13]	[0.06, 0.16]	[0.05, 0.14]	[0.07, 0.19]	[0.05, 0.14]	[0.08, 0.18]
Observations	817	817	817	817	817	817
F Statistic	66.158***	47.772***	80.447***	59.945***	81.873***	59.875***

Notes: This table shows the results of using Genetic-Pareto (GEPA) prompt optimization to design the prompt used when making the LLM-derived predictions. Specifically, summaries are created using Opus 4.5 and then the sample from Table 3 is split based on the median date into "Early" and "Late" sub-samples. The Early sub-sample is used as a training set to which we apply GEPA to develop a prompt that minimizes one of two loss functions: (i) the mean squared error (MSE) or (ii) MSE minus the covariance between the actual and predicted $r_{i,t}^{abn}$. Using the Late sub-sample as an evaluation set, we present regressions of abnormal returns ($r_{i,t}^{abn}$) on the earnings surprise and the LLM-derived predictions ($\hat{g}_{p^*}(\omega_{i,t}, \Omega_{t-})$) under the optimized prompt p^*. Standard errors are presented in parentheses under coefficient estimates. 99% confidence intervals based on 1,000 bootstrap samples are presented in brackets under estimates.

Table 7: Agentic Context Engineering

	Dependent variable: $r_{i,t}^{abn}$			
	Baseline		With ACE Rulebook	
	(1)	(2)	(3)	(4)
α	0.195***	0.118***	0.199***	0.128***
	(0.031)	(0.033)	(0.027)	(0.029)
$\hat{g}_{p^*}(\omega_{i,t}, \Omega_{t-})$	0.502***	0.481***	0.506***	0.484***
	(0.052)	(0.051)	(0.045)	(0.044)
$s_{i,t}$		0.175***		0.166***
		(0.029)		(0.028)
R^2	0.103	0.141	0.137	0.171
	[0.06, 0.16]	[0.09, 0.20]	[0.08, 0.20]	[0.12, 0.24]
Adjusted R^2	0.102	0.139	0.136	0.169
	[0.06, 0.16]	[0.09, 0.20]	[0.08, 0.20]	[0.12, 0.24]
Observations	817	817	817	817
F Statistic	93.241***	66.897***	128.986***	84.077***

Notes: This table shows the results of using Agentic Context Engineering (ACE) to develop a "rulebook" used when making the LLM-derived predictions. Specifically, summaries are created using Opus 4.5 and then the sample from Table 3 is split based on the median date into "Early" and "Late" sub-samples. The Early sub-sample is used as a training set to which we apply ACE to learn a rulebook of general insights that improve predictions. Using the Late sub-sample as an evaluation set, we present regressions of abnormal returns ($r_{i,t}^{abn}$) on the earnings surprise and the LLM-derived predictions ($\hat{g}_{p^*}(\omega_{i,t}, \Omega_{t-})$) under the optimized context p^*. Standard errors are presented in parentheses under coefficient estimates. 99% confidence intervals based on 1,000 bootstrap samples are presented in brackets under estimates.

Table 8: Effect of Model Size and Reasoning Effort

	Dependent variable: $r_{i,t}^{abn}$			
		With ACE		
	(1) Opus 4.5 (High)	(2) Haiku 4.5	(3) Sonnet 4.5	(4) Opus 4.5 (Low)
α	0.118***	0.186***	0.238***	0.195***
	(0.033)	(0.030)	(0.025)	(0.027)
$\hat{g}_{p^*}(\omega_{i,t}, \Omega_{t-})$	0.481***	0.380***	0.323***	0.417***
	(0.051)	(0.047)	(0.039)	(0.045)
$s_{i,t}$	0.175***	0.172***	0.174***	0.164***
	(0.029)	(0.029)	(0.029)	(0.029)
R^2	0.141	0.117	0.121	0.139
	[0.09, 0.20]	[0.07, 0.18]	[0.07, 0.18]	[0.08, 0.21]
Adjusted R^2	0.139	0.115	0.119	0.137
	[0.09, 0.20]	[0.06, 0.18]	[0.07, 0.18]	[0.08, 0.20]
Observations	817	817	817	817
F Statistic	66.897***	54.119***	56.091***	65.714***

Notes: This table shows the interaction between using a rulebook developed using Agentic Context Engineering (ACE) and model size or reasoning effort. Specifically, summaries are created using Opus 4.5 and then the sample from Table 3 is split based on the median date into "Early" and "Late" sub-samples. The Early sub-sample is used as a training set to which we apply ACE to learn a rulebook of general insights that improve predictions. Using the Late sub-sample as an evaluation set, we present regressions of abnormal returns ($r_{i,t}^{abn}$) on the earnings surprise and the LLM-derived predictions ($\hat{g}(\omega_{i,t}, \Omega_{t-})$) under the optimized context p^*. Standard errors are presented in parentheses under coefficient estimates. 99% confidence intervals based on 1,000 bootstrap samples are presented in brackets under estimates.

A Variable Definitions

Table A1: Mathematical Notation and Variable Definitions

Variable	Definition
Time and Indexing	
i	Stock index
t	Effective trading event date (the first trading day on which the announcement information is actionable)
t_c	Time of the earnings conference call
d	Calendar date of the earnings event
$\tau(d)$	Next trading day following calendar date d
h	Return-measurement horizon ($h = 1$ day in our implementation)
q	Evaluation period (e.g., a calendar quarter)
$\mathcal{E}$	Benchmark evaluation sample, $\{(i_k, t_k)\}_{k=1}^{N}$
N	Number of discrete information events in $\mathcal{E}$
N_q	Number of valid observations in evaluation period q
Information Sets and Pricing Function	
$\omega_{i,t}$	Content (text and audio) released for stock i at event t
Ω_{t-}	Investors' information set strictly prior to the release
Ω^a_{t-}	Analysts' information set strictly prior to the release ($\Omega^a_{t-} \subseteq \Omega_{t-}$)
$g(\cdot)$	Unobservable market-wide information pricing function
$\hat{g}(\cdot)$	AI system's scalar prediction (a submission to the benchmark)
$\nu_{i,[t,t+h]}$	Idiosyncratic noise, orthogonal to $g(\cdot)$
Returns and Performance	
$r_{i,[t,t+h]}$	Return for stock i over the window $[t, t+h]$
$r_{i,t}$	Daily return for stock i on day t (including dividends)
$r_{m,t}$	Market return on day t (proxied by VTI)
$r^{abn}_{i,t}$	Abnormal return for stock i on day t ($r_{i,t} - r_{m,t}$)
$p_{i,t-}$	Pre-announcement stock price for stock i
$PRank(x_{i,t})$	Cross-sectional percentile rank of variable x within evaluation period q
Earnings and Fundamentals	
$e_{i,t}$	Reported earnings per share for stock i at event date t
$\mathbb{E}[e_{i,t} \mid \Omega^a_{t-}]$	Consensus (mean) analyst forecast of EPS formed prior to the release
$s_{i,t}$	Standardized earnings surprise for stock i at event date t

Continued on next page

Table A1 (continued)

Variable	Definition
Factor Model and Benchmark Regression	
$\boldsymbol{\beta}_{i,t}$	Vector of factor loadings for stock i
$\mathbf{F}_{[t,t+h]}$	Vector of factor realizations over $[t, t+h]$
$\epsilon_{i,[t,t+h]}$	Idiosyncratic return component from the factor model
$\mathbf{X}_{i,t}$	Vector of known, quantitative determinants used as controls (e.g., SUE)
$u_{i,[t,t+h]}$	Residual from the cross-sectional benchmark regression